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Detecting a Known Signal in Additive Gaussian Noise

EEE 554 Lecture #16

Consider the detection problem:

$$H_0 : \underline{X} = \underline{N}$$

$$H_1 : \underline{X} = \underline{S} + \underline{N}$$

where $\underline{S} \in \mathbb{R}^n$ is known, and $\underline{N} \sim \mathcal{N}[0, \sigma^2 I]$ with σ^2 known.

With $P(H_0)$ and $P(H_1)$ known, the Bayesian decision rule gives:

$$\rightarrow H_1 = \left\{ \underline{X} \middle| \frac{f_{\underline{X}}(\underline{X}|H_1)}{f_{\underline{X}}(\underline{X}|H_0)} > \frac{P(H_0)}{P(H_1)} \right\}$$

More generally, the Neyman-Pearson lemma holds that a decision rule of the form:

$$\rightarrow H_1 = \left\{ \underline{X} \middle| \frac{f_{\underline{X}}(\underline{X}|H_1)}{f_{\underline{X}}(\underline{X}|H_0)} > T \right\}$$

will maximize P_d for any fixed P_{fa} . Thus, the likelihood ratio:

$$\Lambda(\underline{X}) = \frac{f_{\underline{X}}(\underline{X}|H_1)}{f_{\underline{X}}(\underline{X}|H_0)}$$

is a crucial quantity in signal detection.

For the problem under consideration:

$$f_{\underline{X}}(\underline{X}|H_0) = C \exp \left\{ -\frac{1}{2\sigma^2} \underline{X}^T \underline{X} \right\}$$

$$f_{\underline{X}}(\underline{X}|H_1) = C \exp \left\{ -\frac{1}{2\sigma^2} (\underline{X} - \underline{S})^T (\underline{X} - \underline{S}) \right\}$$

with $C = (2\pi\sigma^2)^{-n/2}$. Thus,

$$\Lambda(\underline{X}) = \exp \left\{ \frac{1}{2\sigma^2} (\underline{X}^T \underline{S} + \underline{S}^T \underline{X} - \underline{S}^T \underline{S}) \right\}$$

Note that $\underline{X}^T \underline{S} = \underline{S}^T \underline{X}$ is the inner product of $\underline{X}$ with $\underline{S}$, and $\underline{S}^T \underline{S} = \|\underline{S}\|^2$ is the energy of the signal $\underline{S}$. So,

$$\begin{aligned} \rightarrow H_1 &= \{ \Lambda(\underline{X}) > T \} = \left\{ \exp \left\{ \frac{1}{\sigma^2} \underline{X}^T \underline{S} - \frac{1}{2\sigma^2} \|\underline{S}\|^2 \right\} > T \right\} \\ &= \left\{ \frac{1}{\sigma^2} \underline{X}^T \underline{S} - \frac{1}{2\sigma^2} \|\underline{S}\|^2 > \log T \right\} \\ &= \left\{ \underline{X}^T \underline{S} > \sigma^2 \log T + \frac{1}{2} \|\underline{S}\|^2 \right\} = \{ \underline{X}^T \underline{S} > T' \} \end{aligned}$$

where $T' = \sigma^2 \log T + \frac{1}{2} \|\underline{S}\|^2$.

Detector Performance

The detector for a known signal vector $\underline{S}$ in additive white gaussian noise of known variance σ^2 has the form:

$$\rightarrow H_1 = \{\underline{S}^T \underline{X} > T'\} = \{\rho > T'\}$$

with the detection statistic $\rho = \underline{S}^T \underline{X}$.

Under H_0 , $\underline{X} = \underline{N}$ and hence:

$$\rho_N = \underline{S}^T \underline{N} = \sum_{k=1}^n s_k n_k \sim \mathcal{N} \left[0, \sigma^2 \sum_{k=1}^n s_k^2 \right] = \mathcal{N}[0, \sigma^2 \|\underline{S}\|^2]$$

Under H_1 , $\underline{X} = \underline{S} + \underline{N}$ and:

$$\rho_{S+N} = \underline{S}^T (\underline{S} + \underline{N}) = \|\underline{S}\|^2 + \underline{S}^T \underline{N} \sim \mathcal{N} [\|\underline{S}\|^2, \sigma^2 \|\underline{S}\|^2]$$

So the probabilities of false alarm (P_{fa}) and detection (P_d) are:

$$P_{fa} = P(\rho > T' | H_0) = 1 - \Phi \left(\frac{T'}{\sigma \|\underline{S}\|} \right)$$

and

$$P_d = P(\rho > T' | H_1) = 1 - \Phi \left(\frac{T' - \|\underline{S}\|^2}{\sigma \|\underline{S}\|} \right)$$

Complex Signal and Colored Noise

Now consider a known complex signal vector $\underline{S} \in \mathbb{C}^n$ in additive complex gaussian noise $\underline{N} \sim \mathcal{CN}[0, C_N]$ with a known covariance matrix C_N .

The detection problem is:

$$\begin{aligned} H_0 : \underline{X} &= \underline{N} \\ H_1 : \underline{X} &= \underline{S} + \underline{N} \end{aligned}$$

for which the likelihood ratio is:

$$\begin{aligned} \Lambda(\underline{X}) &= \frac{C \exp \{-(\underline{X} - \underline{S})^H C_N^{-1} (\underline{X} - \underline{S})\}}{C \exp \{-\underline{X}^H C_N^{-1} \underline{X}\}} \\ &= \exp \{ \underline{S}^H C_N^{-1} \underline{X} + \underline{X}^H C_N^{-1} \underline{S} - \underline{S}^H C_N^{-1} \underline{S} \} \\ &= \exp \{ 2\text{Re}(\underline{S}^H C_N^{-1} \underline{X}) - \underline{S}^H C_N^{-1} \underline{S} \} \end{aligned}$$

The detector thus has the form:

$$\rightarrow H_1 = \{\rho > T'\}$$

where the detection statistic is:

$$\rho = \text{Re}(\underline{S}^H C_N^{-1} \underline{X})$$

which is distributed as: $* \mathcal{N} \left[0, \frac{\underline{S}^H C_N^{-1} \underline{S}}{2} \right]$ under H_0 $* \mathcal{N} \left[\underline{S}^H C_N^{-1} \underline{S}, \frac{\underline{S}^H C_N^{-1} \underline{S}}{2} \right]$ under H_1